

SUPPLEMENTARY MATERIAL FOR "LOCATION MATTERS: LAYER-WISE PROMPT CONDITIONING FOR VOICE-STYLE DISENTANGLEMENT IN TTS"

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1. DATA

We use LibriTTS-R [1] audio with the speaker-level annotations of LibriTTS-P [2]: 515.1 hours of speech from 2295 speakers across 331,141 samples, after length filtering (2443 speakers before). Each speaker carries three independent annotations, each a list of keywords from the vocabulary in Table 1, optionally modified by *slightly* or *very*. Because annotations describe the speaker rather than the utterance, a keyword such as *old* applies to every sample of that speaker. Style features are therefore measured per sample from the audio (Section 2), while voice features are derived per speaker from the annotations (Section 3) except for the pitch, which is measured from audio.

2. FEATURE EXTRACTION

Code and analysis outputs referenced below are in the supplementary materials repository¹ under `supplementary/`, pinned at commit 4b10df1. File names link directly.

¹https://github.com/dcze4/ICASSP_location_matters

2.1. Style features

- **Speaking rate** is the syllable rate over voiced regions, estimated from the energy envelope with a fixed detection threshold (`style.py`).
- **Loudness** is computed from RMS energy over 25 ms frames with a 10 ms hop, converted to dB. Frames below the 10th percentile are discarded as silence; we keep the mean and standard deviation of the remainder as loudness and dynamic variability.
- **Pitch variation** uses Penn [3] with a 10 ms hop, $f_{\min} = 30$ Hz, $f_{\max} = 1000$ Hz and Viterbi decoding. Frames with periodicity below 0.5 are discarded. Remaining F_0 values are converted to semitones relative to the utterance mean, $12 \log_2(F_0/\bar{F}_0)$, so the measure reflects variation *within* an utterance rather than the speaker’s absolute pitch.
- **Arousal and valence** come from the Odyssey model [4] (`emotion.py`). The model also predicts dominance, which we discard: it correlates strongly with arousal and adds no independent conditioning dimension.

2.2. Voice features

Pitch is the only voice feature measured from audio. Age, sex and texture are derived from the annotations. Per speaker, we take the ten longest samples (all, if fewer), extract F_0 as above, filter by periodicity without converting to relative semitones, and take the mean of the per-utterance median F_0 , giving the speaker’s absolute pitch(`v4_pitch.py`).

Pitch categories are assigned *within* annotated sex: quartile boundaries are computed separately for each sex category, so *high pitch* denotes a speaker in the upper quartile relative to others of the same annotated sex rather than an absolute F_0 threshold (`v4_pitch.py`). This matches how pitch is perceived, relative to the expected range of a voice type, but it makes the pitch token conditional on the sex token: *male, high pitch* and *female, high pitch* span different absolute F_0 ranges.

2.3. Discretization

Continuous features are binned into three categories at the 25th and 75th percentiles over the full dataset, so the middle category spans the interquartile range (`s1_categorize.py`). Two features depart from this. Sex uses fixed thresholds of ± 0.25 , because its score distribution is bipolar rather than unimodal (Figure 1); pitch percentiles are computed within each annotated sex rather than over the full dataset (Section 2).

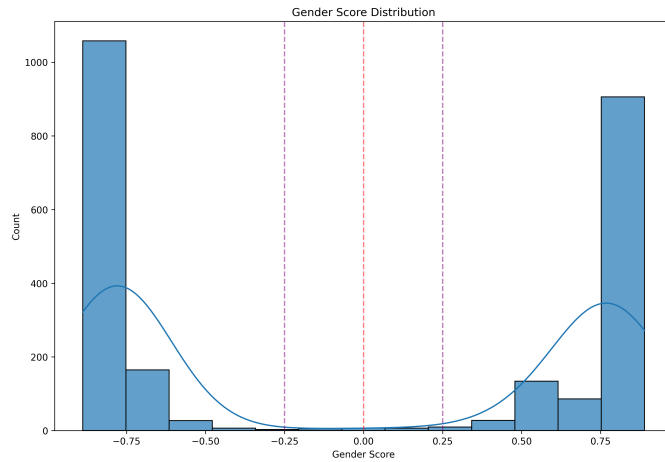


Fig. 1: Sex scores distribution.

3. ANNOTATION SCORING FUNCTION

Annotators differ in how many keywords they use, so a speaker described with many keywords would otherwise contribute more strongly than one described with few. We score each annotator separately, normalize by the number of keywords that

Table 1: Annotation vocabulary grouped by prompt association and category.

Assoc.	Category	Annotations
Voice	Gender	masculine, feminine, gender-neutral
	Age	young, adult-like, middle-aged, mature, old
	Texture	raspy, <i>nasal</i> , <i>muffled</i> , clear, bright, dark, thick, thin, soft, hard, <i>sharp</i>
Style	Arousal	calm, relaxed, tensed, intense, lively, wild
	Valence	friendly, kind, sweet, reassuring, refreshing, sincere, strict
	Speaking rate	fluent, halting
	Loudness	powerful, weak
Ambig.	–	cute, sexy, elegant, intellectual, cool, modest, unique, light

Only Voice annotations are used for prompt creation, and *italic* annotations discarded

annotator used, and average over the three. Every score below lies in $[-1, 1]$, with 0 meaning neutral or unmarked. Writing A_k for the keyword set of annotator k , the speaker score is

$$s = \frac{1}{3} \sum_{k=1}^3 c(A_k), \quad (1)$$

with the per-annotator contribution $c(\cdot)$ defined per feature below.

3.1. Age

Keywords sit on an ordinal scale: *young* = -2 , *adult-like* = -1 , *middle-aged* = 0 , *mature* = $+1$, *old* = $+2$. An intensity modifier shifts a keyword along its own direction: *very* by $+0.5$, *slightly* by -0.5 . So a keyword reaches at most 2.5 in magnitude:

$$c(A) = \frac{1}{2.5 |A|} \sum_{a \in A} w(a). \quad (2)$$

Thus, *very old* scores 1.0, *old* 0.8, and *slightly old* 0.6. Since *middle-aged* has weight 0 and its direction is taken as positive, *very middle-aged* scores $+0.2$ and *slightly middle-aged* -0.2 ; this is deliberate, and unit tests in `v1_age_score.py` measure the behavior.

3.2. Sex

Sex uses a weighted mean rather than a plain one, because the neutral keywords score 0 and would otherwise be indistinguishable from one another. Each keyword has a score $v(a)$ and a weight $\omega(a)$: masculine and feminine take $v \in \{\pm 1, \pm 2, \pm 3\}$ for *slightly*, unmodified and *very*, all with $\omega = 3$; the three gender-neutral variants take $v = 0$ with $\omega = 1, 2, 3$. Then

$$c(A) = \frac{\sum_{a \in A} v(a)}{\sum_{a \in A} \omega(a)}. \quad (3)$$

A *very gender-neutral* judgment therefore pulls a speaker toward 0 more strongly than a *slightly gender-neutral* one (`v2_gender_score.py`).

3.3. Texture

Texture is scored independently on four polar pairs: brightness (bright–dark), thickness (thin–thick), raspiness (clear–raspy) and hardness (soft–hard). For a pair (p^-, p^+) only keywords containing one of the poles are considered; a pole contributes ∓ 1 and the modifiers again shift by ± 0.5 , so the maximum magnitude is 1.5:

$$c(A) = \frac{1}{1.5 |A_p|} \sum_{a \in A_p} w(a), \quad A_p = \{a \in A : p^- \in a \vee p^+ \in a\}. \quad (4)$$

The remaining texture keywords in Table 1 (*nasal*, *muffled* and *sharp*) have no polar partner and are discarded. (`v3_texture_score.py`). There are at most four texture values present per speaker in the dataset. Figure 2b show the distribution.

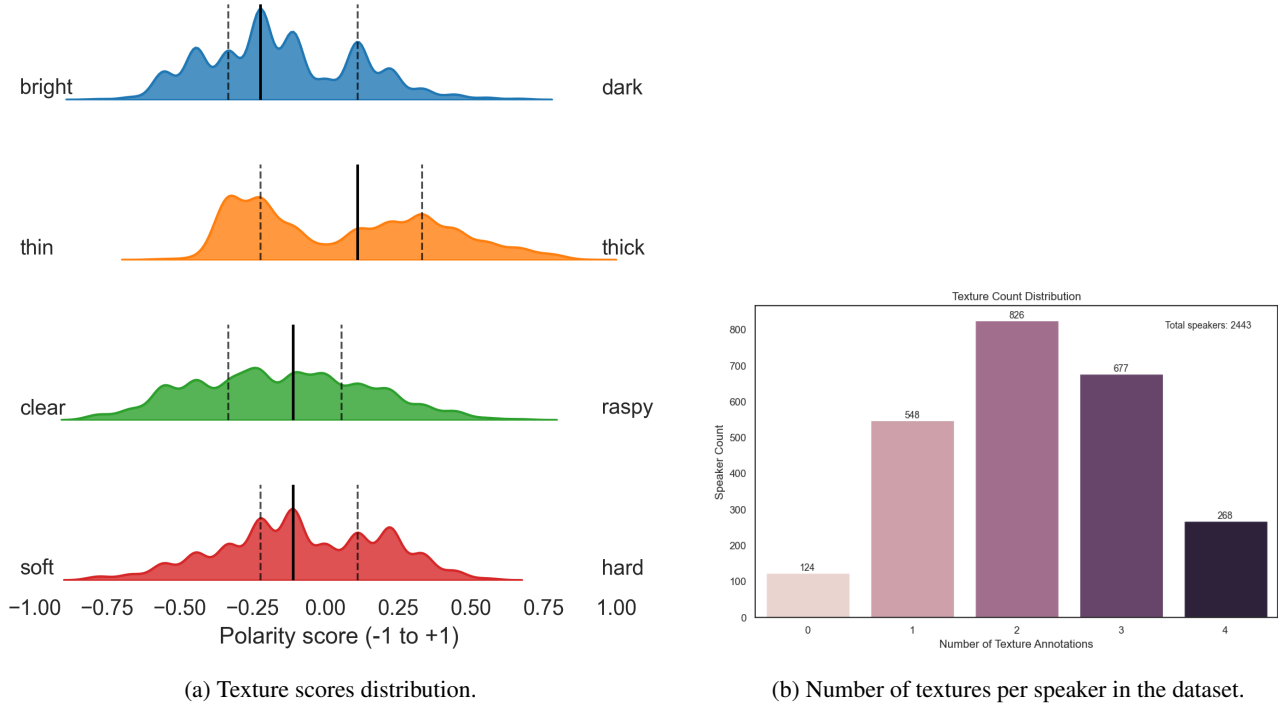


Fig. 2: Texture score distributions and annotation coverage.

Table 2: Prompt feature space.

Prompt	Feature	Possible Values
Voice	Age	young, adult, old
	Sex	male, female, gender-neutral
	Pitch	low, medium, high
	Texture	bright/dark, thin/thick, clear/raspy, soft/hard
Style	Speed	slow, normal, fast
	Loudness	quiet, normal, loud
	Arousal	low, neutral, high
	Valence	negative, neutral, positive

4. PROMPT TEMPLATES

Discretized features (Table 2) are rendered into prompt text in two ways (`prompt_generation.py`).

Structured prompts concatenate feature values in fixed order behind a type label. Word order and function words are identical for every prompt, so the only thing that varies is the feature values. This isolates the effect of conditioning location from any lexical variation.

Template prompts render the same values through handwritten sentence templates: 15 for voice and 20 for style. The voice templates come in four variants covering zero to three texture keywords, so the sentence stays grammatical however many texture features a speaker carries; a speaker with more than three has three sampled at random. During training a template is drawn uniformly at random per sample, so the model sees each feature combination under many surface forms. Table 3 shows examples for both prompting strategies.

Our evaluation aims to isolate the location effects. Therefore, we limit the number to one texture feature when creating the evaluation prompts. Four style features with three values each give $3^4 = 81$ style prompts. Voice prompts combine three three-valued features with at most one texture feature for evaluation, giving $3^3 \times 9 = 243$. During training, however, we allow up to 3 texture features. Figure ?? shows the distribution over speakers.

Table 3: Prompting strategy examples. Each row pair shows structured and template prompts for the same feature combination. Template prompts are randomized during training; two examples are shown per type to illustrate lexical diversity.

Type	Strategy	Prompt
Voice	Structured	Voice: old, female, high pitch, and bright texture.
	Template (id:2)	An old female voice featuring high pitch and bright timbre.
	Template (id:14)	High-pitched and bright, this female voice sounds old.
Voice	Structured	Voice: young, male, low pitch.
	Template (id:8)	Male speaker who is young, displaying low pitch.
	Template (id:15)	The voice presents as male and young, with low pitch.
Style	Structured	Style: fast speed, loud, high arousal, positive valence.
	Template (id:2)	Speaking quickly and loudly with high arousal and positive sentiment.
	Template (id:19)	The speech carries high arousal and positive tone, moving quickly and loudly.
Style	Structured	Style: normal speed, normal loudness, neutral arousal, neutral valence.
	Template (id:13)	Neutral arousal, neutral valence delivery at normal speed, spoken at normal volume.
	Template (id:15)	It is said at normal speed with neutral arousal and neutral valence, keeping it at normal volume.

5. AGGREGATE METRICS ACROSS EMBEDDING SPACES

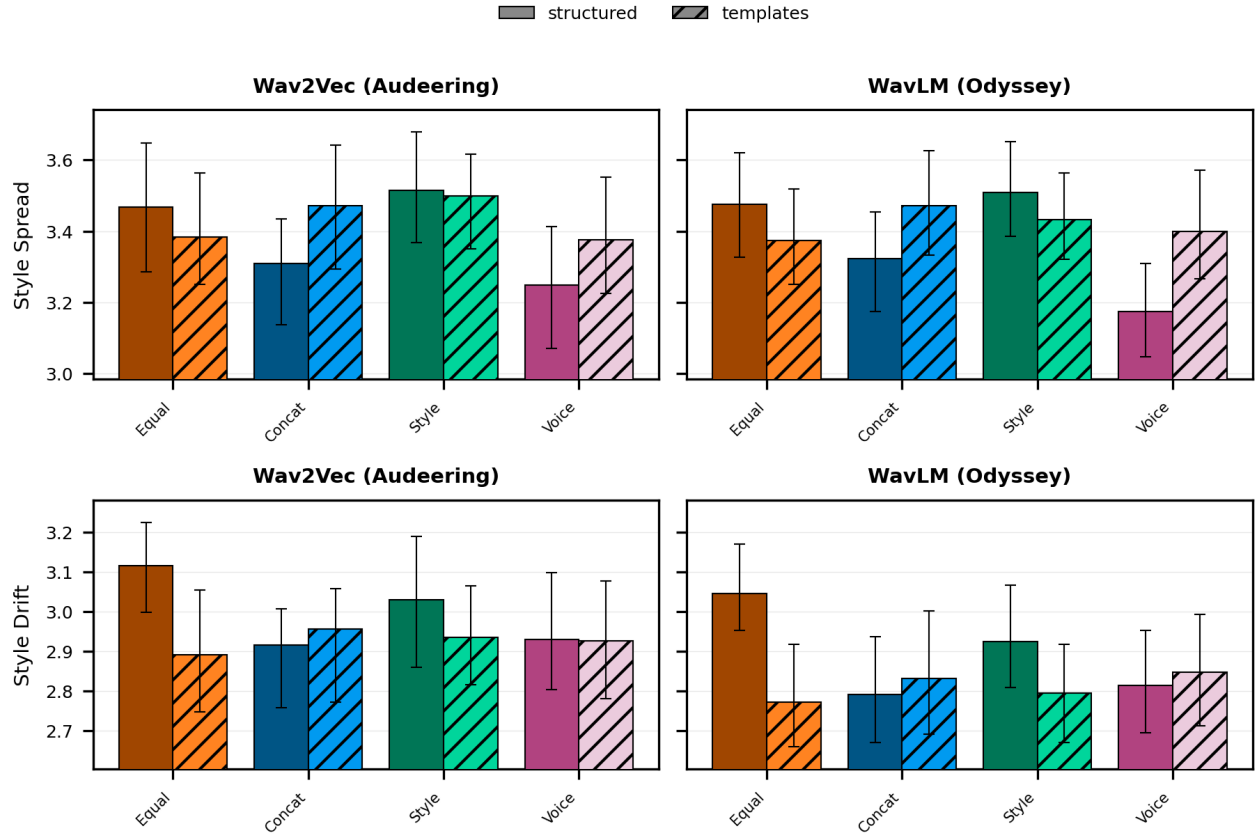
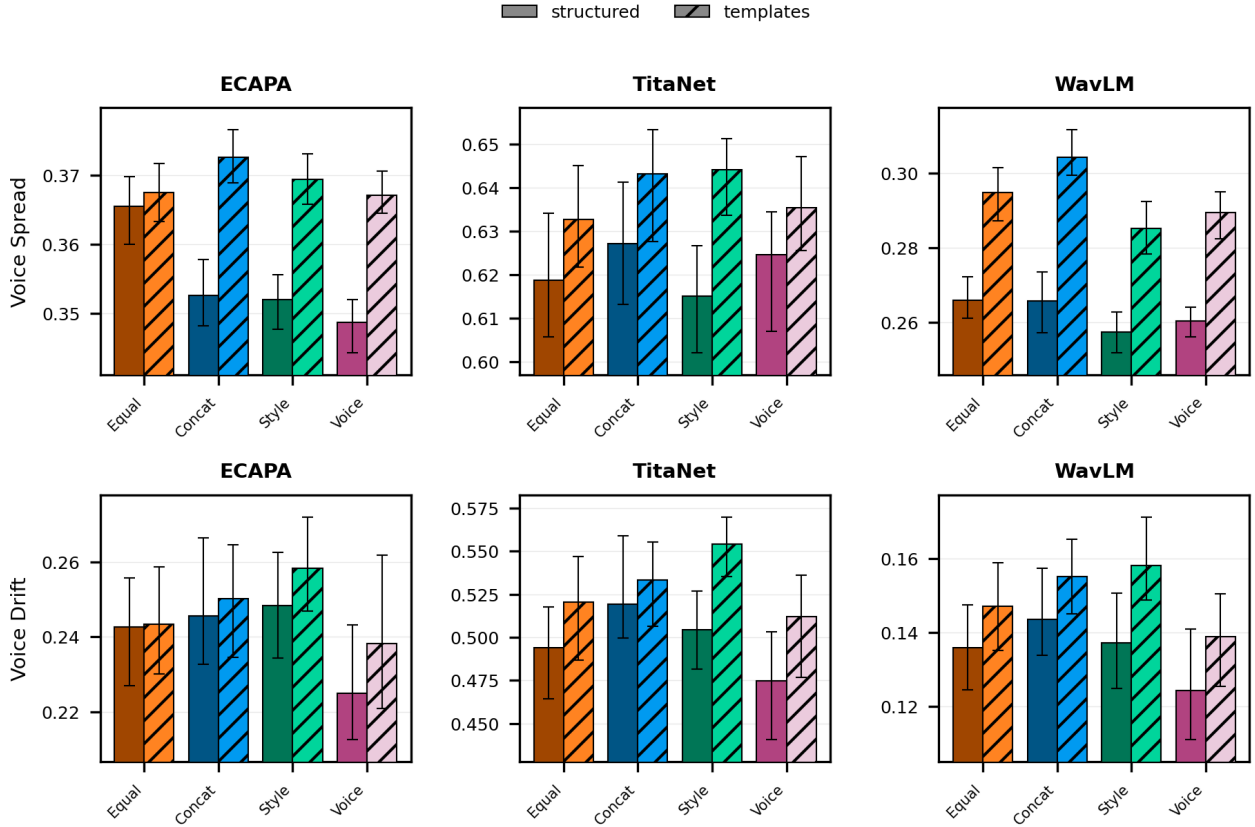


Fig. 3: Style spread ($\uparrow$) and style drift ($\downarrow$) per conditioning strategy, structured (solid) vs. template (hatched) prompting, in two style embedding spaces. Bars are medians; error bars the interquartile range.

Table 4: Style Per-Dimension GMD on WavLM-Odyssey embeddings

Feature	Concat		Style		Voice		Equal	
	Spread $\uparrow$	Drift $\downarrow$	Spread $\uparrow$	Drift $\downarrow$	Spread $\uparrow$	Drift $\downarrow$	Spread $\uparrow$	Drift $\downarrow$
Syl. Rate	1.085	0.984	1.138	0.977	1.072	0.974	1.133	1.018
Energy μ	1.027	0.555	1.178	0.656	1.010	0.592	1.147	0.818
Energy std	0.812	0.734	1.018	0.912	0.934	0.862	1.197	1.155
Pitch std	1.072	1.033	1.133	1.078	1.027	1.024	1.039	0.996
Pitch Range	1.079	1.046	1.119	1.073	1.051	1.054	1.040	1.006
Arousal	1.177	0.780	1.180	0.769	0.973	0.732	1.065	0.769
Valence	1.117	0.989	1.098	0.947	0.997	0.928	1.071	0.956

**Fig. 4:** Voice spread ($\uparrow$) and voice drift ($\downarrow$) per conditioning strategy, structured (solid) vs. template (hatched) prompting, in three speaker embedding spaces. Bars are medians; error bars the interquartile range.

Figures 3 and 4 report spread and drift for every conditioning strategy under both prompting regimes, in each proxy embedding space.

6. PER-DIMENSION ANALYSIS

Table 4 reports the Gini Mean Difference [5] (GMD) for spread and drift per dimension on WavLM-Odyssey embeddings. Unlike variance, GMD does not square differences and thus it is more robust to outliers.

The trade-off observed in aggregate metrics holds at the per-feature level. Style-early achieves the highest spread on 5 out of 7 dimensions, while Voice-early is the worst at spread on 5 out of 7 dimensions. Voice-early has the lowest drift on 3 out of 7 dimensions, particularly on the emotion dimensions, suggesting that emotion features may be more tightly coupled with voice space.

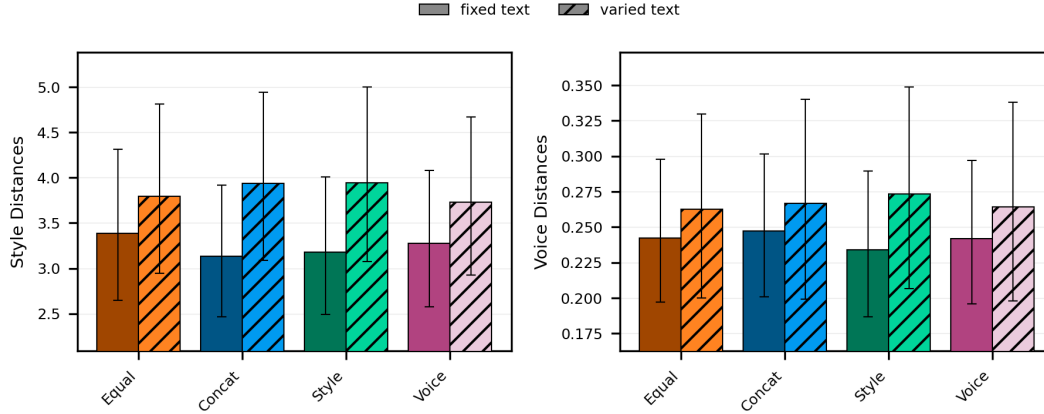


Fig. 5: Varying text effect on style and voice pairwise distances

identity (in this dataset) than acoustic-prosodic features.

Conclusion: Table 4 confirms that early placement of style conditioning improves expressiveness across most individual features, not just in aggregate, and that the spread-drift trade-off (discussed in main paper subsection 5.1) operates at feature level.

7. EFFECT OF INPUT TEXT

We investigate how input text affects voice and style embeddings independent of prompt conditioning. Both prompts are fixed, and the input text is varied over 210 sentences from EmergentTTSEval [6] (Emotions, Questions, Syntactic Complexity) at the first evolution depth. We report mean pairwise distance as Style and Voice Distance. Text carries fine-grained prosodic cues such as punctuation, question marks, and syntactic complexity. These cues encode style information independently of the style prompt. The fixed-text condition serves as baseline, capturing the model’s inherent variation when all inputs are held constant. Fig. 5 confirms encoded prosodic cues. With varying text, style pairwise distances reach and even exceed the style spread observed in prompt-varying experiments (Main paper subsection 5.1). However, this analysis uses neutral prompts, which also correspond to the most frequent category in the training data and therefore impose weaker constraints on the model’s output. Under stronger conditioning, text-driven variation would likely be more limited. Voice distances also increase but remain close to the voice drift range, suggesting that text primarily affects style rather than voice identity.

Conclusion: The input text is not a neutral carrier but a competing style signal. Under weak prompt conditioning, text-driven prosodic cues account for substantial style variation.

8. FULL MIXEDLM/GEE OUTPUT

Full MixedLM and GEE output is provided as plain text together with the script that produced it: `structured_reanalysis.txt` and `templates_reanalysis.txt` for structured and template prompting respectively, both fitted by `mixedlm_reanalysis.py`.

9. REFERENCES

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